



Attempt this Paper on this Question Sheet only.

OBJECTIVE TYPE

Q1. Read the following items carefully and encircle the correct option listed below at each item.
(One mark for each)

- i) The F-Ratio in a completely randomized ANOVA is:
a) MST/MSE b) $MSE/MSTR$ c) MSE/MST d) $MSTR/MSE$
- ii) An ANOVA procedure is applied to data obtained from 5 samples, where each sample contains 9 observations. The degrees of freedom for the critical value of F are:
a) 5 numerator and 9 denominator degrees of freedom
b) 4 numerator and 8 denominator degrees of freedom
c) 45 degrees of freedom
d) 4 numerator and 40 denominator degrees of freedom
- iii) In the ANOVA, treatment refers to.
a) experimental units c) different levels of a factor
b) a factor d) None of these
- iv) The assumptions under analysis of variance consist of:
a) Normality and Independence c) Linearity and Additivity
b) Both (a) and (b) d) None of these
- v) The following design provides the maximum number of degrees of freedom for error sum of squares:
a) Completely Randomized Design c) Completely Randomized Block Design
b) Latin Square Design d) None of these
- vi) Multiple comparisons tests are applicable when:
a) Null Hypothesis about equality of means is rejected
b) Null Hypothesis about equality of means is accepted
c) Does not depend upon the rejection or acceptance of Null Hypothesis
d) None of these
- vii) One can estimate the missing observation through covariance technique by simply changing the sign of....
a) b b) r c) Correction Factor d) None of these
- viii) The efficiency of two experimental designs can simply be measured through of error variances.
a) Addition b) Subtraction c) Multiplication d) Ratio
- ix) Two Latin squares are if each letter of one square design occurs exactly once with every letter of the other square when they are superimposed.
a) Orthogonal b) Factorial Designs c) Efficient d) None of these
- x) A contrast iscombination of treatments.
a) Linear b) Exponential c) Quadratic d) None of these



UNIVERSITY OF THE PUNJAB

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PAPER: Design and Analysis of Experiments (Theory) TIME ALLOWED: 2 hrs. & 30 mins.
Course Code: STAT-303 MAX. MARKS: 50

Attempt this Paper on Separate Answer Sheet provided.

SUBJECTIVE TYPE

SHORT QUESTIONS

Q2. Differentiate between the following terms: (4 marks each)

- i) Random Effects and Mixed Effects
- ii) Contrast and Orthogonal Contrasts
- iii) Completely Randomized and Completely Randomized Block Designs
- iv) Replication and Repetition
- v) Latin Square and Graeco Latin Square Designs.

Q3. a) For a RCB experiment

$$Y_{ij} = \mu + \alpha_i + \beta_j + \varepsilon_{ij} \quad \begin{matrix} i = 1, 2, \dots, r \\ j = 1, 2, \dots, k \end{matrix}$$

Let μ , α_i and β_j are least square estimates of μ , α_i and β_j respectively. Develop expected Mean Squares indicating the assumptions.

- b) An agriculturist approaches you with five varieties of a crop to be compared for their yields in five replicates on a piece of land which varies in fertility from South to North and in water level from East to West. Suggest a suitable layout plan with proper randomization. (6+4)

Q4. a) Seven treatments arranged in six randomized complete blocks gave the following sum of squares and products

S.O.V.	YY	XY	XX
Blocks	1200	600	200
Treatments	800	300	100
Error	1400	700	600

- i) Is the regression of Y on X significance at 5% level.
- ii) Construct ANOVA and write the inference.

- b) Five varieties of Rice are compared in five randomized complete blocks. The SS for Total, Blocks and Varieties are 282.30, 26.96, and 155.44 respectively.

- (i) Test the significance of difference between varietal means
- (ii) Compare the efficiency of this design with the design in which blocks are ignored. (6+4)

Q5. Derive formula for estimating two missing observations in a LS Design when values are missing in different rows, different columns and different treatments. (10)